A Python-Based Numerical Implementation of the One-Dimensional Richards Equation for Soil Water Flow with Root Water Uptake

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Abstract

Understanding water movement in the unsaturated zone is essential for hydrological modelling, irrigation management, and ecohydrological studies. This paper presents a numerical implementation of a one–dimensional soil water flow model based on the Richards equation. The model integrates soil hydraulic functions using the Van Genuchten–Mualem formulation and includes a root water uptake component based on the Feddes stress response model. The implementation is written in Python and uses a modular structure allowing flexible grid specification, solver configuration, and post-processing of hydrological variables. The framework provides an extensible platform for simulating soil moisture dynamics under varying climatic and soil conditions.

1 Introduction

Water movement in unsaturated soils plays a critical role in the hydrological cycle, influencing groundwater recharge, plant water availability, and surface–subsurface interactions. The Richards equation is widely used to describe variably saturated flow in soils. However, solving this equation numerically requires constitutive relationships for soil hydraulic properties and suitable boundary conditions.

This study presents a Python implementation of a one–dimensional Richards equation model that includes:

- Soil hydraulic relationships using the Van Genuchten model
- Root water uptake using the Feddes stress response approach
- Flexible vertical grid discretization
- Time-dependent boundary forcing

The model architecture is designed for research applications and transparent reproducibility.

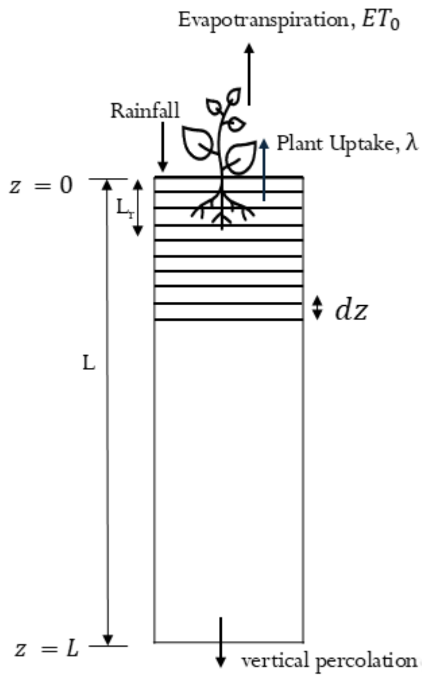

Figure 1: Structure of the soil water flow model. The framework integrates soil hydraulic functions, root water uptake processes, and a numerical Richards equation solver.

2 Governing Equation

Water movement in variably saturated soils is described by the Richards equation [1, 2]:

$$\frac{\partial \theta}{\partial t} = \frac{\partial}{\partial z}\left[K(h)\left(\frac{\partial h}{\partial z} - 1\right)\right] - \lambda \tag{1}$$

where

- h is the pressure head (m)
- $K(h)$ is the hydraulic conductivity (m day^{-1})
- λ is the root water uptake term
- z is vertical depth (positive downward)

3 Soil Hydraulic Properties

The soil hydraulic functions follow the Van Genuchten–Mualem formulation[3].

3.1 Effective Saturation

$$S_e = \left(1 + |\alpha h|^N\right)^{-m} \tag{2}$$

with

$$m = 1 - \frac{1}{N} \tag{3}$$

3.2 Water Content

$$\theta = (\theta_s - \theta_r)S_e + \theta_r \tag{4}$$

where

- θ_s = saturated water content
- θ_r = residual water content

3.3 Hydraulic Conductivity

Hydraulic conductivity is described using the Mualem model:

$$K = K_{sat} S_e^{\eta} \left[1 - \left(1 - S_e^{1/m}\right)^m\right]^2 \tag{5}$$

These relationships are implemented in the hydraulic model module.

4 Root Water Uptake

The plant root water uptake is represented using the Feddes stress response model[4].

4.1 Root Distribution

Root density decreases exponentially with depth:

$$f_1(z) = \frac{a}{L_r} \frac{exp(-a) - exp(-az/L_r)}{(1+a)exp(-a) - 1} \tag{6}$$

where

- L_r is the maximum rooting depth
- a controls root density decay

4.2 Water Stress Function

The water stress response function is defined as

$$f_2(h) = \begin{cases} 0.01 & h \geq h_a \\ 1 & h_a > h > h_d \\ 1 - \frac{h - h_d}{h_w - h_d} & h_d \geq h \geq h_w \\ 0 & h < h_w \end{cases}$$

where

- h_a is the anaerobiosis threshold
- h_d is the soil moisture stress threshold
- h_w is the wilting point

4.3 Root Water Uptake Term, λ

The root water uptake is computed as

$$\lambda = f_1(z)f_2(h)ET_0 \tag{7}$$

where ET_0 is the potential evapotranspiration rate.

5 Numerical Implementation

5.1 Spatial Discretization

The soil profile is discretized using a uniform vertical grid:

$$z_i = z_{min} + \frac{\Delta z}{2} + (i-1)\Delta z \tag{8}$$

where

- Δz is the grid spacing
- i denotes the node index

5.2 Initial Conditions

Initial pressure head is assumed hydrostatic:

$$h(z) = z - z_{wt} \tag{9}$$

where z_{wt} is the water table depth.

5.3 Temporal Integration

The Richards equation is solved using the `solve_ivp` routine from SciPy. Implicit solvers such as the Backward Differentiation Formula (BDF) are used due to the stiffness of the governing equation.

6 Model Outputs

The simulation produces time-dependent profiles of:

- soil water content (θ)
- pressure head (h)
- hydraulic conductivity (K)
- effective saturation (S_e)
- root water uptake
- evapotranspiration
- soil water storage, (Θ)
- runoff generation

These outputs enables evaluation of soil moisture dynamics and plant–soil interactions.

7 Implementation

The model is implemented in Python and organized into modular components:

- Soil hydraulic functions
- Grid and simulation configuration classes
- Root water uptake module
- Richards equation solver
- Post-processing utilities

This modular architecture improves reproducibility and facilitates model extension.

8 Results

The code is used to predict soil moisture and pressure head at Johnstown, Wexford in Ireland. The soil texture data is shown below.

Table 1: Soil texture at different depths, Johnstown Castle – Diamond and Sills dataset

Site	Depth min (m)	Depth max (m)	Soil type	Sand (%)	Silt (%)	Clay (%)
Johnstown Castle	0.00	0.25	loam	49	32	19
	0.25	0.60	loam	49	33	18
	0.60	0.90	loam	36	47	17
	0.90	1.20	loam	51	36	13

Table 2: Mean (μ) and standard deviation (σ) of VG parameters at different depths for Diamond and Sills dataset at Johnstown Castle [5]

Depth (m)		θ_s		θ_r		$\log_{10}(\alpha)$		$\log_{10}(n)$		$\log_{10}(K_{sat})$		η	
Min	Max	μ	σ	μ	σ	μ	σ	μ	σ	μ	σ	μ	σ
0.00	0.25	0.394	0.006	0.085	0.005	-2.004	0.075	0.142	0.012	1.156	0.081	-0.676	1.118
0.25	0.60	0.394	0.006	0.083	0.005	-2.013	0.074	0.144	0.012	1.184	0.081	-0.623	1.098
0.60	0.90	0.409	0.008	0.082	0.006	-2.294	0.075	0.168	0.013	1.247	0.126	-0.010	1.343
0.90	1.20	0.393	0.007	0.071	0.006	-2.015	0.074	0.155	0.011	1.366	0.096	-0.447	0.983

The results are shown below.

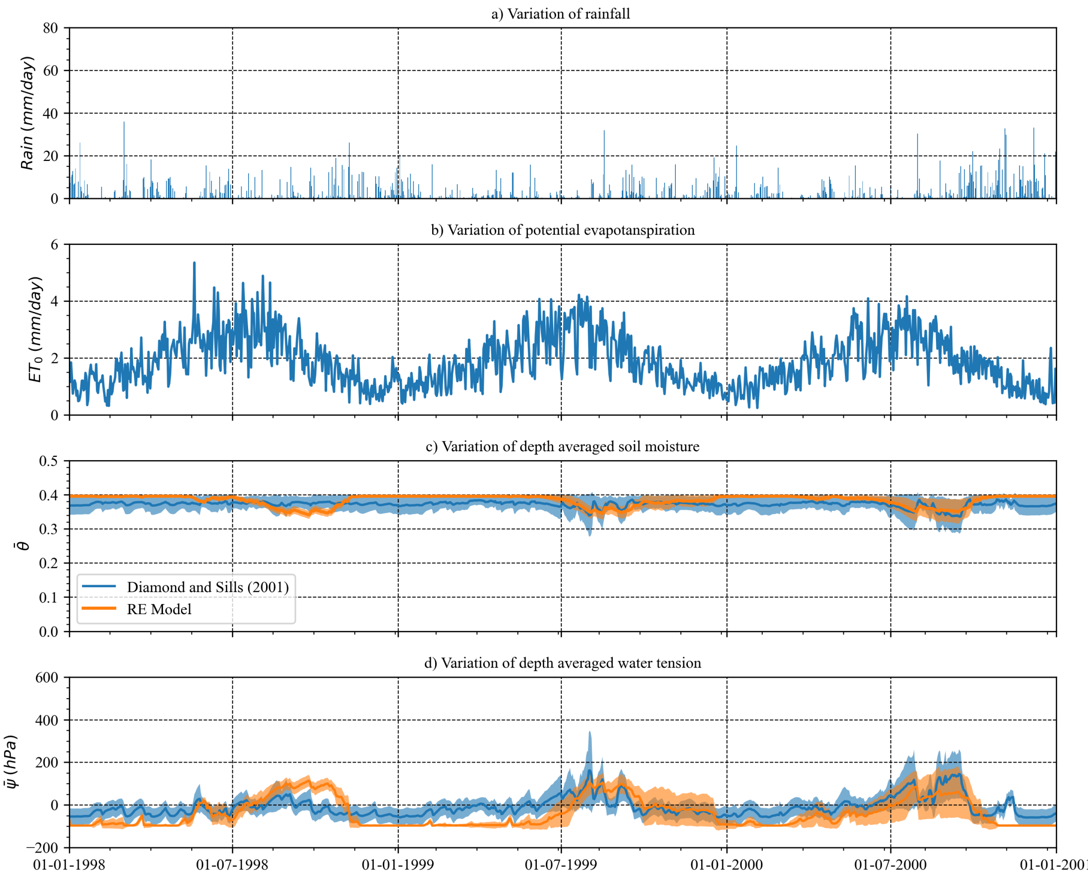

Figure 2: Plot of a) daily rainfall (mm/day), b) potential evapotranspiration (ET_0) (mm/day), c) depth averaged soil moisture ($\bar{\theta}$), and d) depth averaged water tension ($\bar{\psi}$) time series for Diamond and Sills dataset at Johnstown Castle.

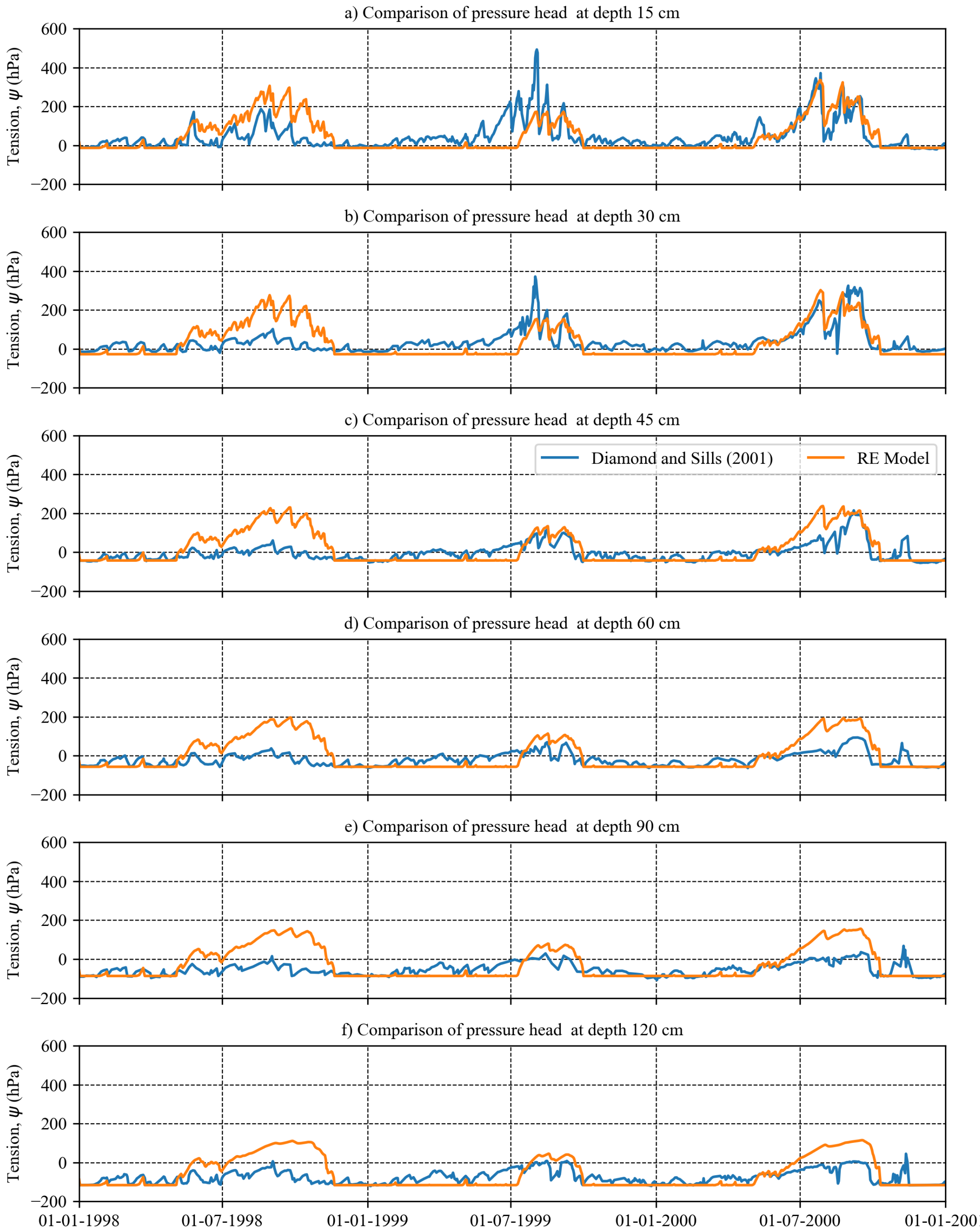

Figure 3: Comparison of predicted water tension (hPa) values at different depths against observed water tension (hPa) values at Johnstown Castle in Diamond and Sills dataset.

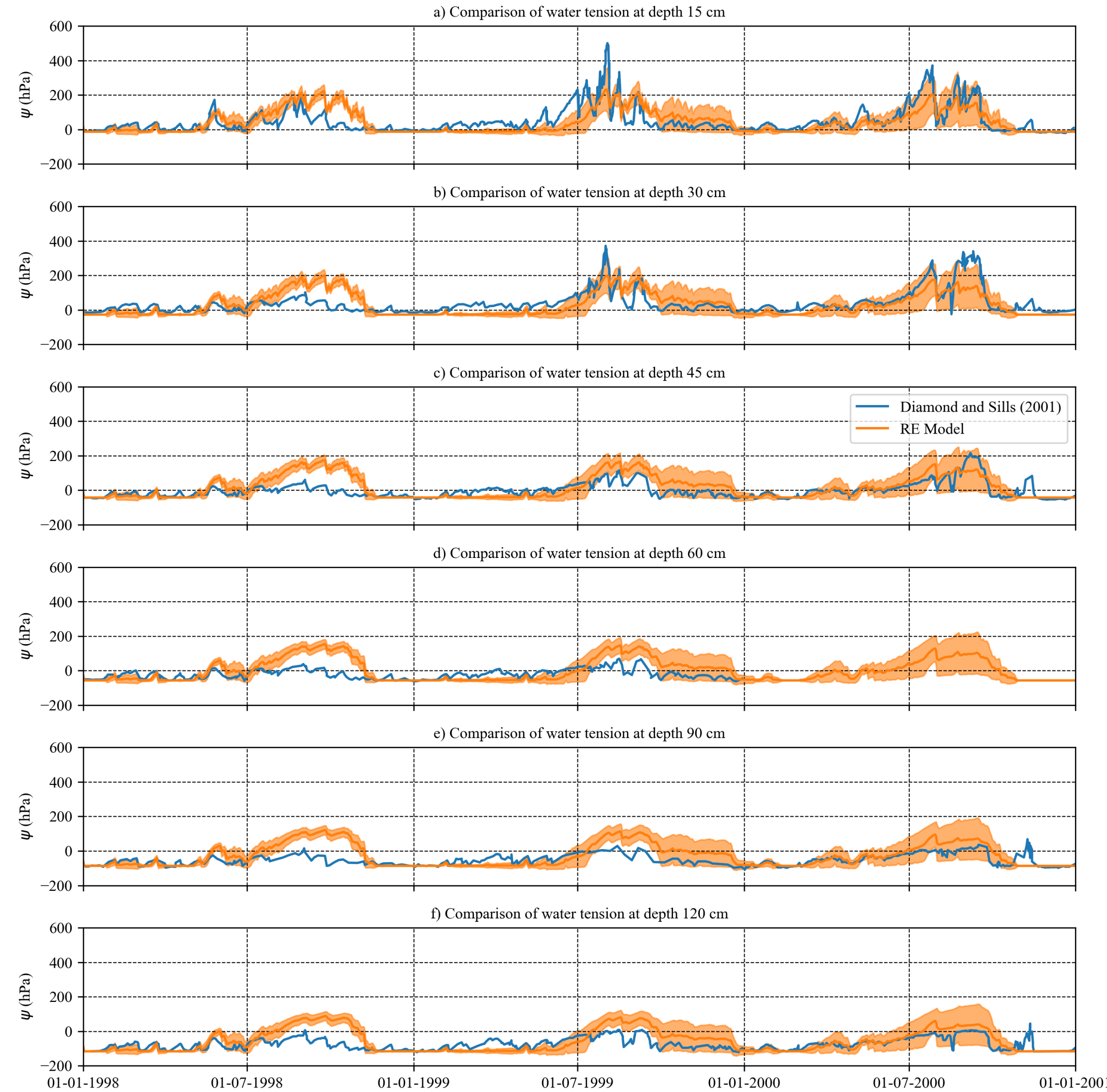

Figure 4: Comparison of predicted water tension (hPa) values at different depths against observed water tension (hPa) values at Johnstown Castle in Diamond and Sills dataset in a Monte Carlo framework.

9 Conclusions

The framework provides a modular and extensible implementation of the 1D Richards Equation for Irish grassland applications. The models support both deterministic and uncertainty-based simulations and are suitable for hydrological and agricultural research workflows.The framework can be extended to include additional hydrological processes such as layered soils, preferential flow, or coupled surface–subsurface dynamics.

Code Availability

The model source code is available in a public GitHub repository.

References

[1] L. A. Richards. Capillary conduction of liquids through porous mediums. *Physics*, 1(5):318–333, 11 1931.

[2] Michael A. Celia, Elias T. Bouloutas, and Regina L. Zarba. A general mass-conservative numerical solution for the unsaturated flow equation. *Water Resources Research*, 26(7):1483–1496, 1990.

[3] M. Th. van Genuchten. A closed-form equation for predicting the hydraulic conductivity of unsaturated soils. *Soil Science Society of America Journal*, 44(5):892–898, 1980.

[4] Reinder A. Feddes, Piotr Kowalik, Krystina Kolinska-Malinka, and Henryk Zaradny. Simulation of field water uptake by plants using a soil water dependent root extraction function. *Journal of Hydrology*, 31(1):13–26, 1976.

[5] J. Diamond and P. Sills. Soil water regimes. End of project report, Teagasc, Johnstown Castle Research Centre, Wexford, Ireland, June 2001. End of Project Report, 29 pages.